

Chapter 6: Geometrical figures

10 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Measurement and space: MA4-ANG-C-01 (angle relationships, triangles and quadrilaterals) and transformations of shapes on the Cartesian plane and grids.



6.01 Transforming shapes

6.02 Combined transformations

6.03 Line symmetry

6.04 Rotational symmetry

6.05 Classifying triangles

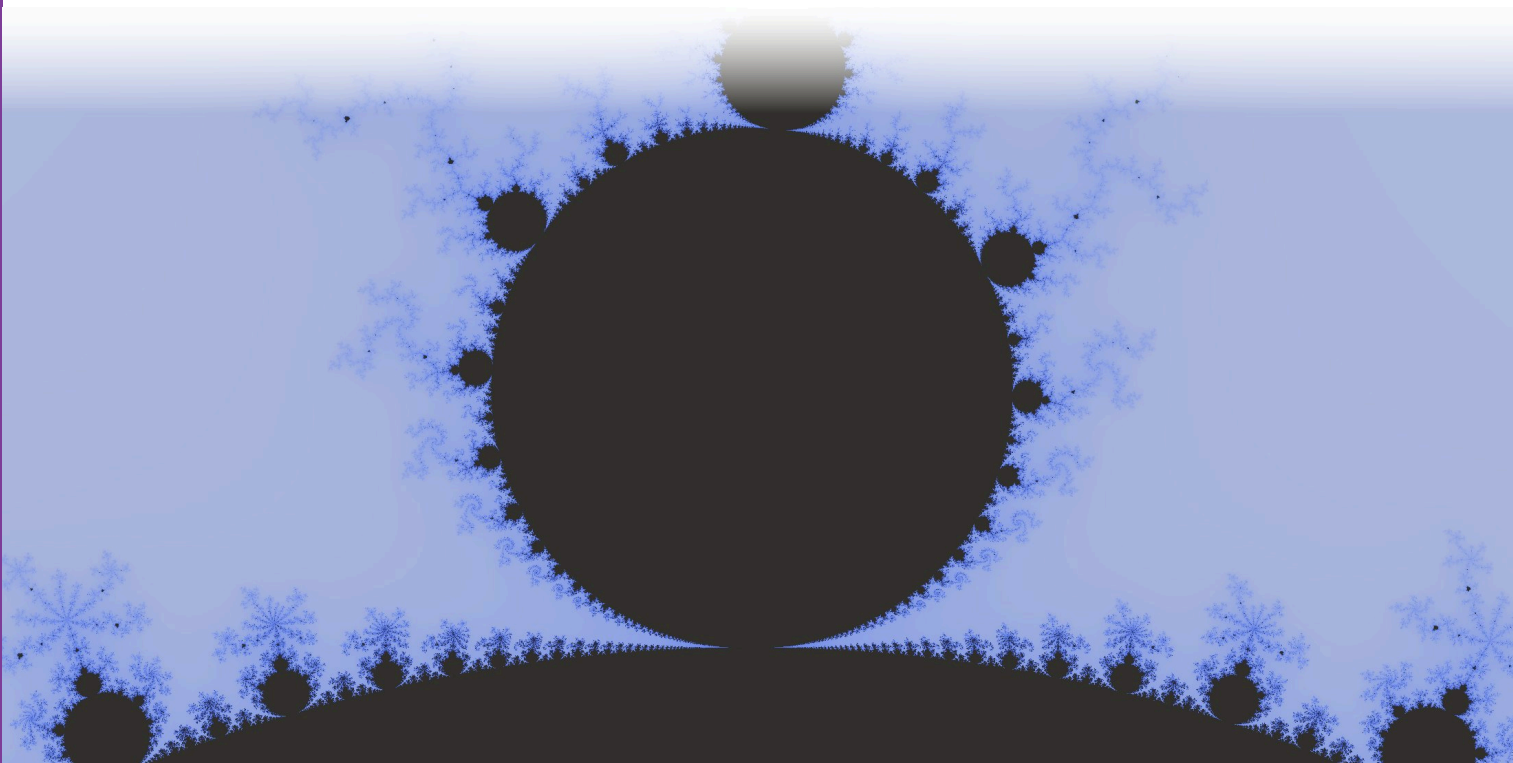
6.06 Angle sum of a triangle

6.07 Exterior angle of a triangle

6.08 Quadrilaterals

6.09 Angle sum of a quadrilateral

6.10 Properties of quadrilaterals



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO

Grey boxes. Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO

Blue boxes. Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL

Skill drill pages. After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

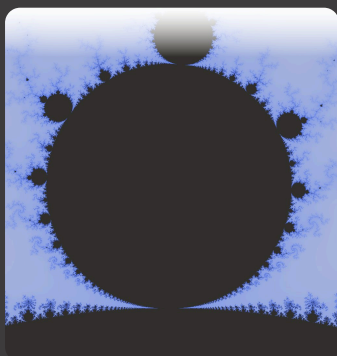
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
6.01	Transforming shapes	3		☐
6.02	Combined transformations	9		☐
6.03	Line symmetry	15		☐
6.04	Rotational symmetry	21		☐
6.05	Classifying triangles	27		☐
6.06	Angle sum of a triangle	33		☐
6.07	Exterior angle of a triangle	39		☐
6.08	Quadrilaterals	45		☐
6.09	Angle sum of a quadrilateral	51		☐
6.10	Properties of quadrilaterals	57		☐

In this chapter you will

- reflect, translate and rotate shapes
- find line and rotational symmetry
- classify triangles and quadrilaterals
- use the angle sums of triangles and quadrilaterals
- use the exterior angle of a triangle
- describe the properties of special quadrilaterals



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the top bulb of the set and the lacy edge around it. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
6.01

Transforming shapes

Date _____

Learning intentions

We are learning to:

- reflect, translate and rotate a shape on a grid

Success criteria

I can:

- ☐ **reflect** a shape across a mirror line
- ☐ **translate** a shape a given distance left, right, up or down
- ☐ **rotate** a shape 90° , 180° or 270° about a point

Key terms Write the definition as your teacher explains it.

Reflect

Translate

Rotate

Notes

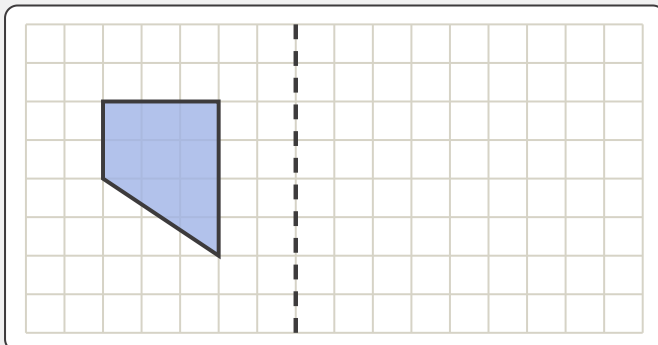
Level 1 · Getting started

Easy · 25 questions

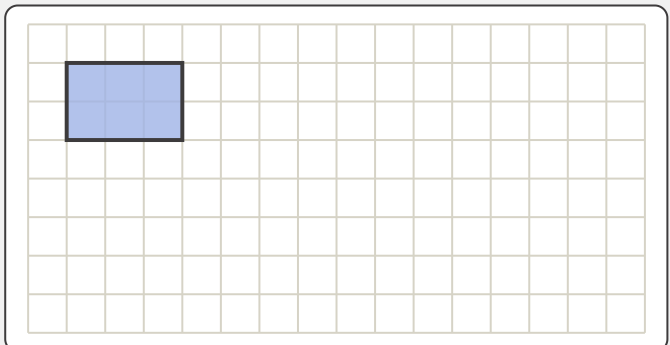
WE DO With your teacher: copy what they write on the board.

Example 1 Reflect, then translate the shape.

a Reflect the shape in the dotted line.



b Translate the shape 5 right and 3 down.



YOU DO

On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. 25 QUESTIONS

REMEMBER REFLECT: flip over the mirror line; same distance each side.

ROUND 1 Questions 1–9: translate (slide) the point. Write where it ends up.

1 $(-1, -6)$: 1 left, 3 down

2 $(2, -5)$: 1 right, 2 down

3 $(6, -2)$: 4 right

4 $(-3, -4)$: 3 right

5 $(2, 0)$: 6 right, 6 down

6 $(4, -2)$: 5 left, 1 up

7 $(3, 5)$: 1 left, 2 up

8 $(0, 3)$: 1 left, 2 up

9 $(-3, 4)$: 3 left, 4 down

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 10–25: reflect (flip) the point in the axis given.

10 $(-2, 6)$ in the x-axis

11 $(-2, 1)$ in the x-axis

12 $(-6, 2)$ in the x-axis

13 $(8, 6)$ in the x-axis

14 $(2, -3)$ in the x-axis

15 $(-7, -6)$ in the y-axis

16 $(-8, -4)$ in the y-axis

17 $(-3, 1)$ in the x-axis

18 $(-5, 8)$ in the x-axis

19 $(-4, 1)$ in the x-axis

20 $(3, -2)$ in the x-axis

21 $(-1, -2)$ in the x-axis

22 $(-6, 2)$ in the y-axis

23 $(6, -6)$ in the x-axis

24 $(-7, 4)$ in the y-axis

25 $(6, -4)$ in the x-axis

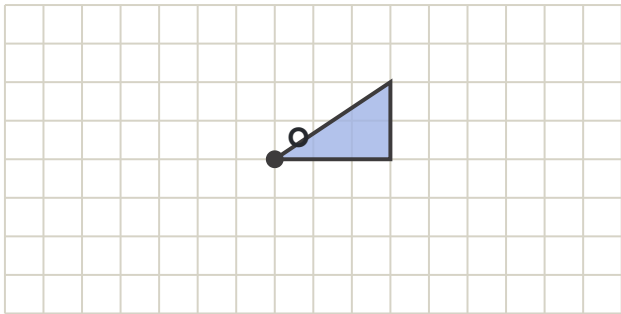
Level 2 · Building up

Medium · 15 questions

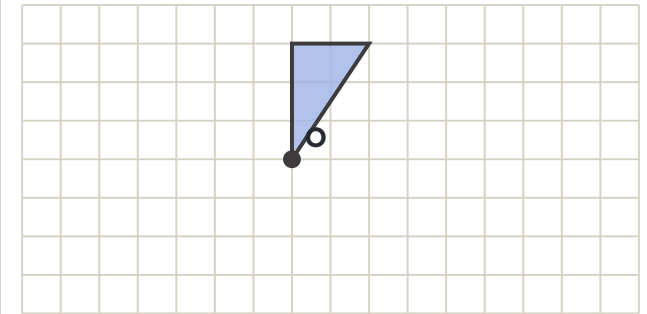
WE DO With your teacher: copy what they write on the board.

Example 2 Rotate the shape about O.

a Rotate 180° about O.



b Rotate 90° clockwise about O.



YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

15 QUESTIONS

REMEMBER TRANSLATE: slide; count units left/right then up/down.

ROUND 1 Questions 1–15: rotate (turn) the point about the origin.

1 (1, 4), 90° clockwise

2 (–2, –5), 90° clockwise

3 (–2, 6), 180°

4 (4, 5), 180°

5 (6, 6), 90° anticlockwise

6 (–3, –4), 90° clockwise

7 (–5, 5), 90° clockwise

8 (–6, –3), 180°

9 (–4, –4), 90° clockwise

10 (2, 4), 90° clockwise

11 (4, 1), 180°

12 (–6, –4), 180°

13 (5, –2), 90° clockwise

14 (–6, –1), 180°

15 (–3, –6), 180°

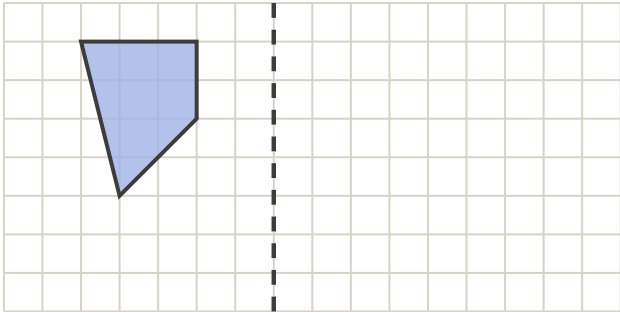
Level 3 · Challenge

Challenging · 2 questions · then try the extension

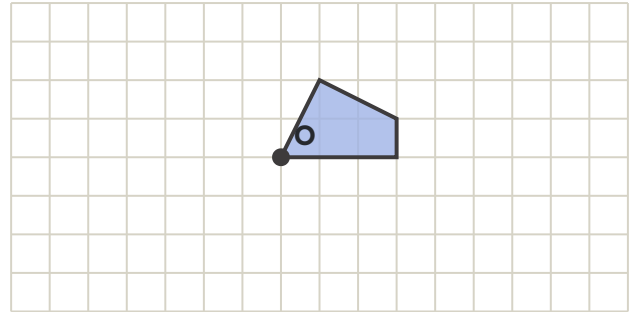
WE DO With your teacher: copy what they write on the board.

Example 3 Reflect, then rotate the shape.

a Reflect in the dotted line.



b Rotate 270° clockwise about O.

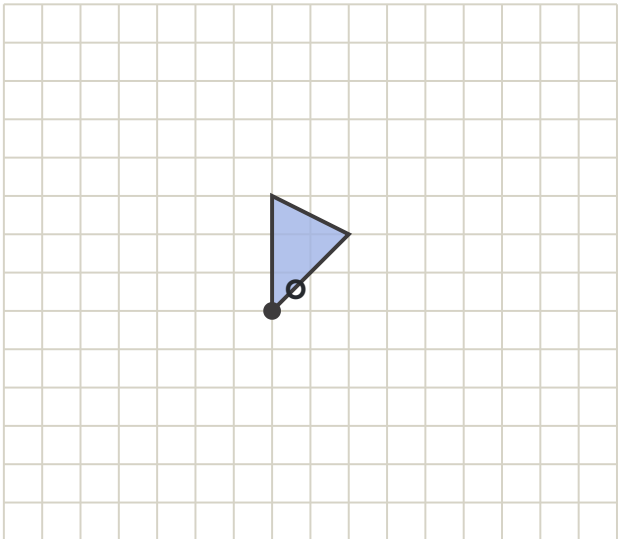


YOU DO On your own: **Challenging** Transform each shape.

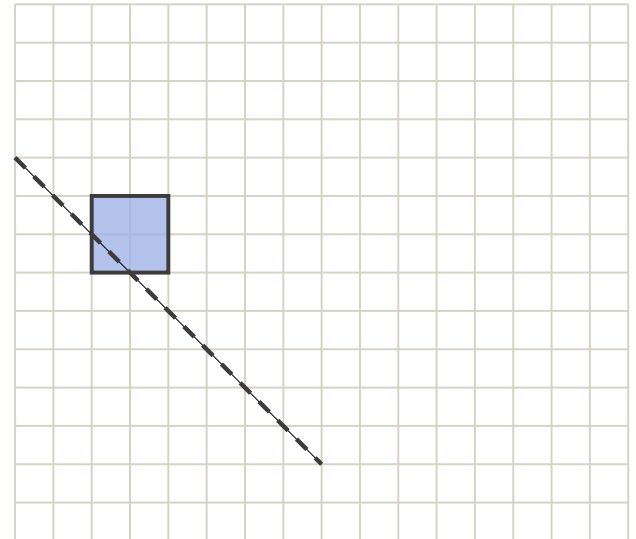
2 QUESTIONS

REMEMBER **ROTATE**: turn about a point; use tracing paper.

1 Rotate 270° clockwise about O.



2 Reflect in the dotted line.



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

A shape is rotated 90° clockwise and then 180° .
What single rotation does the same thing?

1. Rotations about the same point add
2. $90^\circ + 180^\circ = 270^\circ$

Answer: 270° clockwise (or 90° anticlockwise)

YOU DO

On your own: read the worked example, then try these.

- E1 Which single transformation is the same as reflecting in a vertical line and then in a horizontal line that crosses it?

- E2 Is a 90° clockwise rotation the same as a 270° anticlockwise rotation? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Name the transformation that flips a shape.

Level 2

A point is at (2, 3) on a grid. Where is it after a translation 4 right and 1 down?

Level 3

Describe what a 90° clockwise rotation does to an arrow pointing up.

Lesson summary

Missed the lesson? Start here.

- Remember**
- 1. Reflect:** flip over the mirror line; same distance each side.
 - 2. Translate:** slide; count units left/right then up/down.
 - 3. Rotate:** turn about a point; use tracing paper.
 - 4.** The shape keeps its size and shape.

Level 1

Reflect in a vertical line

1. Each vertex the same distance on the other side

Answer: a mirror image

Level 2

Translate 5 right, 3 down

1. Move every vertex

Answer: same shape, new position

Level 3

Rotate 180° about O

1. Each vertex goes through O to the other side

Answer: upside down

Extension

90° then 180°

1. Add

Answer: 270° clockwise

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.02

Combined transformations

Date _____

Learning intentions

We are learning to:

- apply two or more transformations in order
- describe a combined transformation

Success criteria

I can:

- ☐ do each transformation in the order given
- ☐ label each new image (A, A', A'')
- ☐ describe a combination that maps one shape onto another

Key terms Write the definition as your teacher explains it.

Image

Combined transformation

Notes

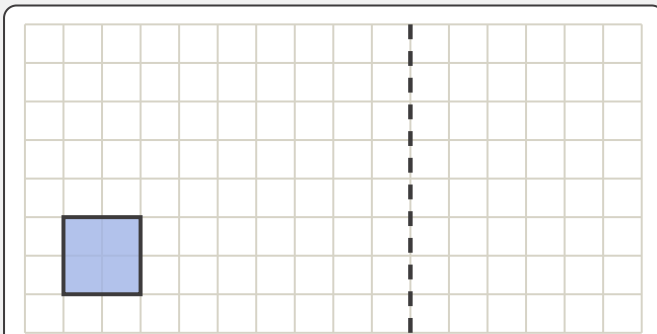
Level 1 · Getting started

Easy · 21 questions

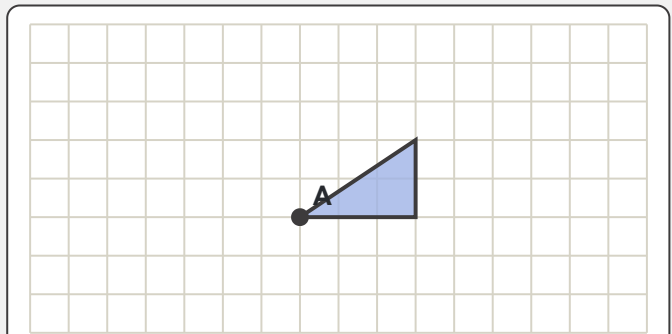
WE DO With your teacher: copy what they write on the board.

Example 1 Do both transformations in order.

a Translate 5 right, 2 up; then reflect.



b Rotate 90° clockwise about A, then 3 left.



YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. 21 QUESTIONS

REMEMBER Do the transformations in the order given.

ROUND 1 Questions 1–6: name the transformation: translation, reflection or rotation.

1 a flip

2 a butterfly's two wings

3 moving 5 units up

4 a clock hand moving

5 a door opening

6 a turn

YOU DO

On your own: **Easy** keep going: Round 2.

15 QUESTIONS MORE

ROUND 2

Questions 7–21: two translations. Write the single translation that does the same.

7 7 right, then 5 left

8 7 left, then 5 left

9 6 right, then 4 right

10 1 right, then 3 right

11 7 right, then 7 right

12 6 left, then 2 right

13 5 right, then 3 left

14 7 right, then 1 left

15 7 right, then 2 left

16 6 right, then 5 right

17 1 left, then 7 left

18 4 left, then 3 left

19 4 right, then 3 left

20 4 left, then 2 left

21 5 left, then 7 left

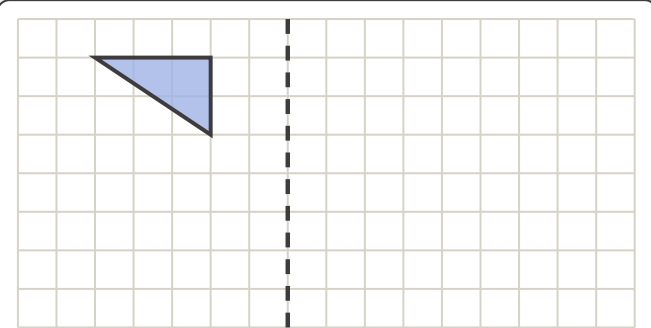
Level 2 · Building up

Medium · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Do both transformations, then compare.

a Reflect in the dotted line, then translate 1 down.



b Does the order matter? Try reflecting then translating versus translating then reflecting.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

12 QUESTIONS

REMEMBER Draw each image before doing the next.

ROUND 1 Questions 1–12: do both transformations in order. Where does the point end up?

1 $(-5, -5)$: 1 right, then reflect in the y-axis

2 $(4, 3)$: 4 right, then reflect in the x-axis

3 $(-5, 5)$: 1 right, then reflect in the y-axis

4 $(1, 4)$: 1 right, then reflect in the y-axis

5 $(-1, 1)$: 5 right, then reflect in the x-axis

6 $(-2, 5)$: 4 right, then reflect in the y-axis

7 $(2, 5)$: 5 right, then reflect in the x-axis

8 $(-1, 5)$: 5 right, then reflect in the y-axis

9 $(-5, 5)$: 5 right, then reflect in the y-axis

10 $(-3, 4)$: 1 right, then reflect in the x-axis

11 $(3, -2)$: 2 right, then reflect in the y-axis

12 $(2, 3)$: 3 right, then reflect in the y-axis

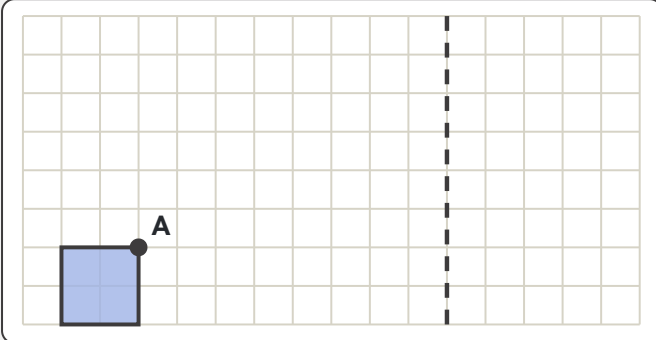
Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Do all three transformations, then describe a combination.

- a Translate 5 right and 2 up, rotate 90° clockwise about A, then reflect in the dotted line.



- b Describe two transformations that move a shape from the top-left to the bottom-right of a grid, upside down.

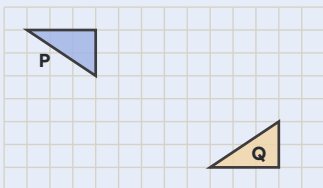
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Describe the transformations.

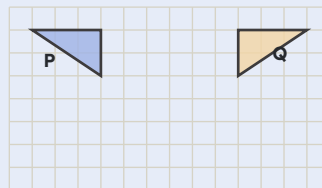
2 QUESTIONS

REMEMBER Label them A, A', A''.

- 1 Describe how shape P moves onto shape Q.



- 2 Find two different ways to move P onto Q.





Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

A triangle has vertices $(1, 1)$, $(3, 1)$, $(3, 4)$. It is reflected in the y -axis, then translated 2 units down. Where are the vertices?

1. Reflect in y -axis: $(-1, 1)$, $(-3, 1)$, $(-3, 4)$
2. Down 2: subtract 2 from each y

Answer: $(-1, -1)$, $(-3, -1)$, $(-3, 2)$

YOU DO

On your own: read the worked example, then try these.

- E1 Reflect $(2, 5)$ in the x -axis, then translate 3 left. Where is it?

- E2 Rotate $(4, 1)$ by 180° about the origin. Where is it?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

What does A' mean?

Level 2

A point $(1, 2)$ is translated 3 right then 4 up. Where is it?

Level 3

A point $(2, 3)$ is reflected in the x -axis then translated 2 left. Where is it?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Do the transformations in the order given.
 2. Draw each image before doing the next.
 3. Label them A, A', A''.
 4. Order can change the final position.

Level 1

Translate then reflect

1. Slide first
2. Then flip over the line

Answer: image A''

Level 2

Rotate then translate

1. Turn about the point
2. Then slide

Answer: image A''

Level 3

Order matters?

1. Try both ways

Answer: often yes

Extension

Reflect in y-axis then down 2

1. $(x, y) \rightarrow (-x, y - 2)$

Answer: $(-1, -1)$, $(-3, -1)$, $(-3, 2)$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.03

Line symmetry

Date _____

Learning intentions

We are learning to:

- find the axes of symmetry of a shape

Success criteria

I can:

- ☐ know that an axis of symmetry divides a shape into two mirror images
- ☐ draw every axis of symmetry on a shape
- ☐ count the axes of symmetry

Key terms Write the definition as your teacher explains it.

Line symmetry

Axis of symmetry

Notes

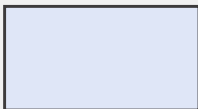
Level 1 · Getting started

Easy · 29 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Draw every axis of symmetry.

a Draw the axes of symmetry.



b Draw the axes of symmetry.



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **29 QUESTIONS**

REMEMBER An **AXIS OF SYMMETRY** folds a shape onto itself.

ROUND 1 Questions 1–9: how many axes (lines) of symmetry?

1 kite

2 parallelogram

3 isosceles trapezium

4 isosceles triangle

5 regular hexagon

6 semicircle

7 scalene triangle

8 square

9 equilateral triangle

YOU DO

On your own: **Easy** keep going: Round 2.

20 QUESTIONS MORE

ROUND 2

Questions 10–29: how many axes of symmetry does each capital letter have?

10 S

11 M

12 O

13 F

14 H

15 D

16 A

17 T

18 E

19 X

20 N

21 K

22 L

23 C

24 B

25 W

26 I

27 V

28 Z

29 U

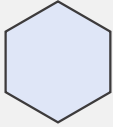
Level 2 · Building up

Medium · 24 questions

WE DO With your teacher: copy what they write on the board.

Example 2 How many axes of symmetry?

a How many axes?



b How many axes?



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

24 QUESTIONS

REMEMBER Test by folding or with a mirror.

ROUND 1 Questions 1–16: how many axes of symmetry does a regular polygon with this many sides have?

1 14 sides

2 6 sides

3 12 sides

4 20 sides

5 16 sides

6 17 sides

7 8 sides

8 13 sides

9 3 sides

10 4 sides

11 19 sides

12 15 sides

13 5 sides

14 10 sides

15 7 sides

16 9 sides

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 17–24: true or false?

17 A scalene triangle has no axes of symmetry.

18 An isosceles triangle has 1 axis of symmetry.

19 A rectangle's diagonals are axes of symmetry.

20 A square has 4 axes of symmetry.

21 A regular octagon has 4 axes of symmetry.

22 A regular hexagon has 6 axes of symmetry.

23 A kite has 1 axis of symmetry.

24 The letter N has an axis of symmetry.

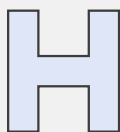
Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 How many axes of symmetry?

a How many axes?



Your teacher works this out. Copy every step.

b How many axes of symmetry does a circle have?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER Draw the axes as dotted lines.

1 How many axes of symmetry does a regular 10-sided shape have?

2 Which capital letters of the alphabet have exactly one axis of symmetry? List 5.

3 Draw a shape with exactly 2 axes of symmetry.

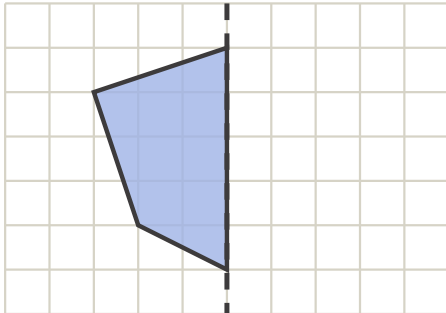
Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

Complete the shape so that the dotted line is an axis of symmetry.

- Each point is reflected across the line
- Join the reflected points

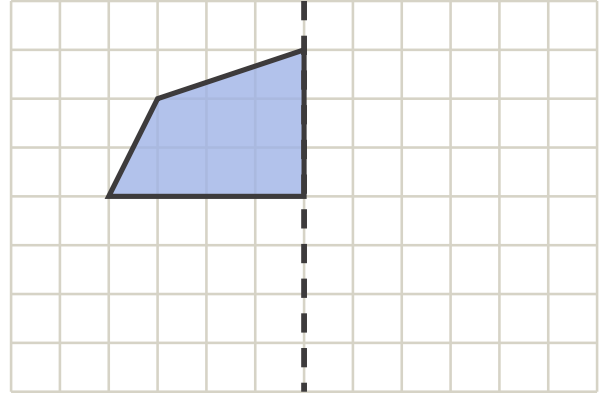


Answer: a symmetric shape

YOU DO

On your own: read the worked example, then try these.

- E1 Complete the shape so both dotted lines are axes of symmetry.



- E2 A regular polygon has 12 axes of symmetry. How many sides does it have?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

How many axes of symmetry does an equilateral triangle have?

Level 2

How many axes of symmetry does a kite have?

Level 3

How many axes of symmetry does a regular hexagon have?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. An **axis of symmetry** folds a shape onto itself.
 2. Test by folding or with a mirror.
 3. Draw the axes as dotted lines.
 4. A regular polygon with n sides has n axes.

Level 1

Rectangle

1. Vertical and horizontal only

Answer: 2 axes

Level 2

Square

1. 2 through sides, 2 diagonals

Answer: 4 axes

Level 3

Parallelogram

1. No fold works

Answer: 0 axes

Extension

Regular 12-gon

1. n axes

Answer: 12 axes

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.04

Rotational symmetry

Date _____

Learning intentions

We are learning to:

- find the order of rotational symmetry of a shape

Success criteria

I can:

- ☐ know a shape has rotational symmetry if it looks the same before a full turn
- ☐ count the **order**: how many times it matches in one full turn
- ☐ find the angle of each turn

Key terms Write the definition as your teacher explains it.

Rotational symmetry

Order

Notes

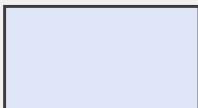
Level 1 · Getting started

Easy · 29 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Give the order of rotational symmetry.

a Order of rotational symmetry?



b Order?



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **29 QUESTIONS**

REMEMBER Turn the shape about its centre.

ROUND 1 Questions 1–9: order of rotational symmetry (write 1 for none).

- | | | |
|-----------------|----------------------|--------------------|
| 1 parallelogram | 2 square | 3 circle |
| 4 rectangle | 5 isosceles triangle | 6 regular pentagon |
| 7 kite | 8 scalene triangle | 9 rhombus |

YOU DO

On your own: **Easy** keep going: Round 2.

20 QUESTIONS MORE

ROUND 2

Questions 10–29: order of rotational symmetry of each capital letter (write 1 for none).

10 U

11 H

12 X

13 F

14 M

15 N

16 C

17 T

18 B

19 E

20 Z

21 V

22 S

23 K

24 I

25 D

26 W

27 O

28 A

29 L

Level 2 · Building up

Medium · 23 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Give the order of rotational symmetry.

a Order?



b Order?



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

23 QUESTIONS

REMEMBER Count how many times it looks the same in one turn.

ROUND 1

Questions 1–15: a regular polygon has this many sides. Write its order of rotational symmetry and the smallest angle of turn.

1 9 sides

2 3 sides

3 36 sides

4 10 sides

5 40 sides

6 18 sides

7 6 sides

8 8 sides

9 24 sides

10 20 sides

11 15 sides

12 45 sides

13 5 sides

14 4 sides

15 12 sides

YOU DO On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2 Questions 16–23: true or false?

16 A regular hexagon has order 6.

17 Every shape has rotational symmetry of at least order 1.

18 A shape with order 2 looks the same upside down.

19 A rectangle has rotational symmetry of order 4.

20 A square has rotational symmetry of order 4.

21 The letter S has rotational symmetry of order 2.

22 An equilateral triangle turns onto itself every 120° .

23 A regular pentagon turns onto itself every 60° .

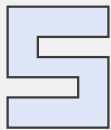
Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Give the order, then find the angle.

a Order?



Your teacher works this out. Copy every step.

b A shape has order 5. What is the smallest angle it turns to match itself?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER That count is the **ORDER**.

1 Which capital letters have rotational symmetry of order 2? List 4.

2 Can a shape have rotational symmetry but no line symmetry? Give an example.

3 A shape matches itself every 72° . What is its order?



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

A regular polygon matches itself every 40° . How many sides does it have?

1. A full turn is 360°
2. $360 \div 40 = 9$

Answer: 9 sides

YOU DO

On your own: read the worked example, then try these.

- E1 A regular polygon matches itself every 30° . How many sides does it have?

- E2 Which has the larger angle of turn: order 6 or order 8? Explain.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is the order of rotational symmetry of a square?

Level 2

Does a kite have rotational symmetry?

Level 3

A shape has order 6. Find its smallest angle of turn.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Turn the shape about its centre.
 2. Count how many times it looks the same in one turn.
 3. That count is the **order**.
 4. Order 1 means no rotational symmetry. Angle = $360^\circ \div \text{order}$.

Level 1

Rectangle

1. Matches at 180° and 360°

Answer: order 2

Level 2

Square

1. Every 90°

Answer: order 4

Level 3

Kite

1. Only at 360°

Answer: no rotational symmetry

Extension

Every 40°

1. $360 \div 40$

Answer: 9 sides

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
6.05

Classifying triangles

Date _____

Learning intentions

We are learning to:

- classify triangles by their sides and by their angles

Success criteria

I can:

- ☐ name triangles by sides: equilateral, isosceles, scalene
- ☐ name triangles by angles: acute-angled, right-angled, obtuse-angled
- ☐ find equal angles in an isosceles triangle

Key terms Write the definition as your teacher explains it.

Isosceles

Scalene

Equilateral

Notes

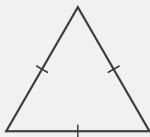
Level 1 · Getting started

Easy · 28 questions

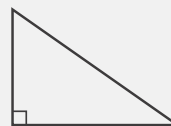
WE DO With your teacher: copy what they write on the board.

Example 1 Classify each triangle by its sides and by its angles.

a Classify by sides and by angles.



b Classify by sides and by angles.



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

28 QUESTIONS

REMEMBER Sides: **EQUILATERAL** (3 equal), **ISOSCELES** (2 equal), **SCALED** (none).

ROUND 1 Questions 1–12: classify by angles: acute, right or obtuse.

1 115°, 42°, 23°

2 37°, 86°, 57°

3 118°, 33°, 29°

4 60°, 76°, 44°

5 64°, 29°, 87°

6 103°, 35°, 42°

7 46°, 53°, 81°

8 52°, 66°, 62°

9 52°, 44°, 84°

10 105°, 51°, 24°

11 50°, 77°, 53°

12 79°, 68°, 33°

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 13–28: classify by sides: equilateral, isosceles or scalene.

13 6 cm, 9 cm, 10 cm

14 4 cm, 4 cm, 4 cm

15 7 cm, 7 cm, 3 cm

16 3 cm, 6 cm, 8 cm

17 7 cm, 7 cm, 6 cm

18 11 cm, 14 cm, 15 cm

19 7 cm, 7 cm, 7 cm

20 8 cm, 8 cm, 4 cm

21 12 cm, 12 cm, 22 cm

22 5 cm, 5 cm, 5 cm

23 12 cm, 12 cm, 12 cm

24 12 cm, 12 cm, 15 cm

25 10 cm, 10 cm, 11 cm

26 12 cm, 12 cm, 4 cm

27 11 cm, 12 cm, 17 cm

28 11 cm, 14 cm, 16 cm

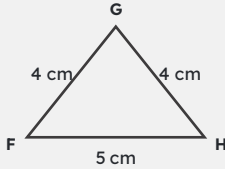
Level 2 · Building up

Medium · 23 questions

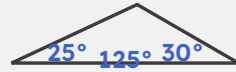
WE DO With your teacher: copy what they write on the board.

Example 2 Classify the triangle, then find the equal angles.

a Classify $\triangle FGH$. Which angles are equal?



b Classify by sides and angles.



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

23 QUESTIONS

REMEMBER Angles: **ACUTE** (all $< 90^\circ$), **RIGHT** (one 90°), **OBTUSE** (one $> 90^\circ$).

ROUND 1 Questions 1–15: classify by sides and by angles.

1 angles $52^\circ, 64^\circ, 64^\circ$

2 angles $60^\circ, 60^\circ, 60^\circ$

3 angles $36^\circ, 72^\circ, 72^\circ$

4 angles $90^\circ, 45^\circ, 45^\circ$

5 angles $125^\circ, 36^\circ, 19^\circ$

6 angles $74^\circ, 53^\circ, 53^\circ$

7 angles $121^\circ, 33^\circ, 26^\circ$

8 angles $48^\circ, 66^\circ, 66^\circ$

9 angles $80^\circ, 50^\circ, 50^\circ$

10 angles $106^\circ, 20^\circ, 54^\circ$

11 angles $124^\circ, 33^\circ, 23^\circ$

12 angles $113^\circ, 20^\circ, 47^\circ$

13 angles $122^\circ, 34^\circ, 24^\circ$

14 angles $116^\circ, 21^\circ, 43^\circ$

15 angles $56^\circ, 62^\circ, 62^\circ$

YOU DO

On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2

Questions 16–23: is it possible to draw this triangle? Write yes or no.

- | | |
|--|---|
| 16 an acute scalene triangle | 17 a triangle with sides 5 cm, 5 cm and 9 cm |
| _____ | _____ |
| 18 an obtuse equilateral triangle | 19 a triangle with two right angles |
| _____ | _____ |
| 20 a triangle with two obtuse angles | 21 an acute isosceles triangle |
| _____ | _____ |
| 22 a triangle with angles 100° , 40° , 40° | 23 a triangle with angles 30° , 60° , 90° |

 Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Classify each triangle from its description.

- a** My angles are 60° , 80° and 40° . Classify me.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Classify each described triangle by sides and by angles.

3 QUESTIONS

REMEMBER Equal sides have matching tick marks.

- 1 My angles are 60° , 80° and 40° .
 - 2 My sides are 5 cm, 5 cm and 7 cm, and one angle is 90° . Is this possible? Explain.
 - 3 My angles are 30° , 30° and 120° .



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

An isosceles triangle has one angle of 40° . What could the other two angles be?

1. If 40° is one of the equal angles: $40^\circ, 40^\circ, 100^\circ$
2. If 40° is the apex: $(180 - 40) \div 2 = 70^\circ$

Answer: 40° and 100° , or 70° and 70°

YOU DO

On your own: read the worked example, then try these.

- E1 An isosceles triangle has one angle of 100° . Find the other two angles.

- E2 An isosceles triangle has one angle of 60° . What type of triangle is it?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Classify a triangle with sides 6, 6 and 6 cm.

Level 2

Classify a triangle with angles 90° , 45° and 45° .

Level 3

Classify a triangle with angles 110° , 35° and 35° by sides and angles.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Sides: **equilateral** (3 equal), **isosceles** (2 equal), **scalene** (none).
 2. Angles: **acute** (all $< 90^\circ$), **right** (one 90°), **obtuse** (one $> 90^\circ$).
 3. Equal sides have matching tick marks.
 4. Angles opposite equal sides are equal.

Level 1

3 equal sides

1. All angles 60°

Answer: equilateral, acute

Level 2

$\triangle FGH$: 4 cm, 4 cm, 5 cm

1. Two equal sides

Answer: isosceles; $\angle F = \angle H$

Level 3

$25^\circ, 30^\circ, 125^\circ$

1. One obtuse angle

Answer: scalene, obtuse

Extension

Isosceles with a 40° angle

1. Two cases

Answer: $40^\circ, 100^\circ$ or $70^\circ, 70^\circ$

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
6.06

Angle sum of a triangle

Date _____

Learning intentions

We are learning to:

- use the angle sum of a triangle to find unknown angles

Success criteria

I can:

- ☐ know the angles in a triangle add to 180°
- ☐ use equal angles in isosceles and equilateral triangles
- ☐ give reasons for each step

Key terms Write the definition as your teacher explains it.

Angle sum

Reason

Notes

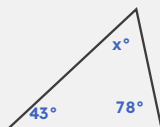
Level 1 · Getting started

Easy · 28 questions

WE DO With your teacher: copy what they write on the board.

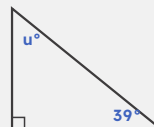
Example 1 Find the unknown angle. Angles in a triangle add to 180° .

a Find x .



Your teacher works this out. Copy every step.

b Find u .



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. 28 QUESTIONS

REMEMBER Angles in a triangle add to 180° .

ROUND 1 Questions 1–12: two angles of a triangle are given. Find the third. (Angles add to 180° .)

- | | | | |
|-----------------------------|------------------------------|------------------------------|-------------------------------|
| 1 68° and 96° | 2 85° and 20° | 3 43° and 53° | 4 46° and 72° |
| 5 29° and 22° | 6 82° and 73° | 7 15° and 32° | 8 112° and 31° |
| 9 85° and 53° | 10 37° and 64° | 11 65° and 76° | 12 24° and 103° |

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 13–28: a right-angled triangle has this angle. Find the other one.

13 11°

14 18°

15 78°

16 74°

17 38°

18 39°

19 10°

20 64°

21 61°

22 57°

23 75°

24 47°

25 28°

26 67°

27 23°

28 48°

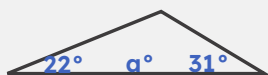
Level 2 · Building up

Medium · 28 questions

WE DO With your teacher: copy what they write on the board.

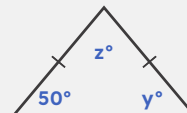
Example 2 Find the unknown angles.

a Find a .



Your teacher works this out. Copy every step.

b Find y and z .



Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

28 QUESTIONS

REMEMBER Add the known angles, subtract from 180° .

ROUND 1 Questions 1–16: isosceles triangle: find the missing angles.

1 apex angle 24°

2 base angle 75°

3 base angle 55°

4 apex angle 50°

5 base angle 25°

6 apex angle 150°

7 base angle 20°

8 base angle 31°

9 base angle 39°

10 base angle 54°

11 base angle 21°

12 base angle 72°

13 base angle 38°

14 base angle 79°

15 base angle 46°

16 apex angle 82°

YOU DO On your own: **Medium** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2 Questions 17–28: the angles of a triangle are given. Find x .

17 $x^\circ, 2x + 15^\circ, 102^\circ$

18 $2x + 20^\circ, x^\circ, 16^\circ$

19 $x^\circ, x + 20^\circ, 90^\circ$

20 $2x + 15^\circ, x^\circ, 117^\circ$

21 $x^\circ, x + 15^\circ, 71^\circ$

22 $x + 20^\circ, 2x^\circ, 19^\circ$

23 $2x + 10^\circ, 2x + 15^\circ, 47^\circ$

24 $x^\circ, 3x^\circ, 120^\circ$

25 $x^\circ, 2x + 20^\circ, 85^\circ$

26 $x + 10^\circ, x^\circ, 42^\circ$

27 $3x^\circ, x^\circ, 48^\circ$

28 $x + 15^\circ, x + 5^\circ, 122^\circ$

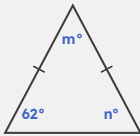
Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find the unknown angles in the isosceles triangles.

a Find n and m (isosceles).



b Find $\angle PQR$.



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Find the unknowns. Give reasons.

3 QUESTIONS

REMEMBER Isosceles: base angles are equal.

1 The angles of a triangle are x° , $2x^\circ$ and $3x^\circ$. Find x .

2 An isosceles triangle has an apex angle of 36° . Find the base angles.

3 Can a triangle have angles of 100° and 90° ? Explain.



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Prove that the angles of a triangle add to 180° using a line parallel to the base.

1. Draw a line through the top vertex parallel to the base
2. The two outer angles equal the base angles (alternate angles)
3. The three angles at the top form a straight line: 180°

Answer: $a + b + c = 180^\circ$

YOU DO

On your own: read the worked example, then try these.

- E1** Tear the corners off a paper triangle and place them together. What do you see?

- E2** Use the angle sum to explain why a triangle can have only one obtuse angle.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Two angles of a triangle are 65° and 45° . Find the third.

Level 2

A right-angled triangle has an angle of 28° . Find the other angle.

Level 3

An isosceles triangle has base angles of 67° . Find the apex angle.

Lesson summary

Missed the lesson? Start here.

Remember

- Angles in a triangle add to **180°**.
- Isosceles: base angles are equal.

- Add the known angles, subtract from 180°.
- Equilateral: every angle is 60°.

Level 1

43°, 78°, x°

1. $180 - 43 - 78$

Answer: $x = 59$

Level 2

Right angle and 39°

1. $180 - 90 - 39$

Answer: $u = 51$

Level 3

Isosceles, apex 72° at P? no: base 72°

1. $\angle PQR = 180 - 72 - 72$

Answer: 36°

Extension

Proof

1. Parallel line and alternate angles

Answer: 180°

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
6.07

Exterior angle of a triangle

Date _____

Learning intentions

We are learning to:

- use the exterior angle property of a triangle

Success criteria

I can:

- ☐ identify an **exterior angle** of a triangle
- ☐ know the exterior angle equals the sum of the two interior opposite angles
- ☐ find unknown angles using this rule

Key terms Write the definition as your teacher explains it.

Exterior angle

Interior opposite angles

Notes

Level 1 · Getting started

Easy · 28 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find the exterior angle.

a Find x .



Your teacher works this out. Copy every step.

b Find x .



Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. 28 QUESTIONS

REMEMBER An **EXTERIOR ANGLE** is made by extending one side.

ROUND 1 Questions 1–12: the two interior opposite angles are given. Find the exterior angle (add them).

- | | | | |
|------------------------|-------------------------|-------------------------|-------------------------|
| 1 74° and 20°
_____ | 2 90° and 34°
_____ | 3 49° and 38°
_____ | 4 17° and 122°
_____ |
| 5 30° and 64°
_____ | 6 32° and 40°
_____ | 7 25° and 63°
_____ | 8 46° and 43°
_____ |
| 9 25° and 55°
_____ | 10 72° and 67°
_____ | 11 68° and 36°
_____ | 12 95° and 39°
_____ |

YOU DO

On your own: **Easy** keep going: Round 2.

16 QUESTIONS MORE

ROUND 2

Questions 13–28: an exterior angle is given. Find the interior angle next to it.

13 143°

14 146°

15 102°

16 75°

17 118°

18 78°

19 49°

20 93°

21 128°

22 66°

23 55°

24 124°

25 20°

26 41°

27 22°

28 44°

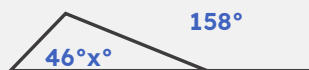
Level 2 · Building up

Medium · 28 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find the unknown interior angle.

a Find x .



Your teacher works this out. Copy every step.

b Find p .



Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

28 QUESTIONS

REMEMBER Exterior angle = sum of the two **INTERIOR OPPOSITE** angles.

ROUND 1 Questions 1–16: the exterior angle and one interior opposite angle are given. Find the other interior opposite angle.

1 exterior 104° , 23°

2 exterior 125° , 32°

3 exterior 137° , 45°

4 exterior 76° , 43°

5 exterior 70° , 45°

6 exterior 107° , 90°

7 exterior 168° , 112°

8 exterior 122° , 68°

9 exterior 113° , 102°

10 exterior 54° , 41°

11 exterior 161° , 18°

12 exterior 102° , 18°

13 exterior 92° , 16°

14 exterior 169° , 39°

15 exterior 163° , 81°

16 exterior 55° , 30°

YOU DO On your own: **Medium** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2 Questions 17–28: find x . The exterior angle equals the sum of the interior opposite angles.

17 exterior 83° , interior opposite 66° and x°

18 exterior 133° , interior opposite 54° and x°

19 exterior 104° , interior opposite 47° and x°

20 exterior 101° , interior opposite 65° and x°

21 exterior 79° , interior opposite 21° and x°

22 exterior 110° , interior opposite 40° and x°

23 exterior 136° , interior opposite 70° and x°

24 exterior 45° , interior opposite 20° and x°

25 exterior 72° , interior opposite 35° and x°

26 exterior 42° , interior opposite 22° and x°

27 exterior 67° , interior opposite 24° and x°

28 exterior 97° , interior opposite 22° and x°

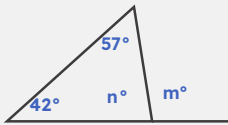
Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find both unknowns, then explain.

a Find m and n .



Your teacher works this out. Copy every step.

b Why is the exterior angle always larger than each interior opposite angle?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Harder problems.

3 QUESTIONS

REMEMBER Exterior + adjacent interior = 180° .

1 The interior opposite angles are x° and $2x^\circ$. The exterior angle is 111° . Find x .

2 The exterior angle of an isosceles triangle at the base is 110° . Find all three interior angles.

3 What do the three exterior angles of a triangle (one at each vertex) add to? Test with an equilateral triangle.



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Prove the exterior angle property using the angle sum.

1. $a + b + c = 180$ (angle sum)
2. $c + x = 180$ (straight line)
3. So $x = a + b$

Answer: The exterior angle equals the sum of the interior opposite angles

YOU DO

On your own: read the worked example, then try these.

- E1** An exterior angle is 4 times the interior angle next to it. Find the exterior angle.

- E2** The exterior angle is 130° and one interior opposite angle is 20° more than the other. Find both.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Interior opposite angles 40° and 75° . Find the exterior angle.

Level 2

The exterior angle is 125° and one interior opposite angle is 60° . Find the other.

Level 3

The exterior angle is 140° . Find the adjacent interior angle.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. An **exterior angle** is made by extending one side.
 2. Exterior angle = sum of the two **interior opposite** angles.
 3. Exterior + adjacent interior = 180° .
 4. Give the reason: "exterior angle of a triangle".

Level 1

59° and 56° opposite

1. $59 + 56$

Answer: $x = 115$

Level 2

$46^\circ + x = 158^\circ$

1. $158 - 46$

Answer: $x = 112$

Level 3

$42^\circ + p = 138^\circ$

1. $138 - 42$

Answer: $p = 96$

Extension

Proof

1. Angle sum and straight line

Answer: $x = a + b$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.08

Quadrilaterals

Date _____

Learning intentions

We are learning to:

- classify quadrilaterals
- describe the symmetry and definitions of special quadrilaterals

Success criteria

I can:

- ☐ decide if a quadrilateral is **convex** or **non-convex**
- ☐ name the special quadrilaterals
- ☐ give the definition of each special quadrilateral

Key terms Write the definition as your teacher explains it.

Convex

Parallelogram

Rhombus

Notes

Level 1 · Getting started

Easy · 16 questions

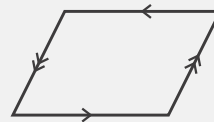
WE DO With your teacher: copy what they write on the board.

Example 1 Convex or non-convex? Name the quadrilateral.

a Convex or non-convex?



b Name this quadrilateral.



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. 16 QUESTIONS

REMEMBER **TRAPEZIUM**: at least one pair of parallel sides. **PARALLELOGRAM**: both pairs parallel.

ROUND 1 Questions 1–6: name the quadrilateral.

1 2 pairs of equal adjacent sides

2 a parallelogram with 4 equal sides

3 a parallelogram with a right angle

4 a trapezium with equal legs

5 2 pairs of parallel sides

6 4 equal sides and 4 right angles

YOU DO On your own: **Easy** keep going: Round 2.

10 QUESTIONS MORE

ROUND 2 Questions 7–16: true or false?

7 Every rectangle is a parallelogram.

8 Every square is a rectangle.

9 Every rhombus is a parallelogram.

10 A quadrilateral has 4 sides.

11 A trapezium has two pairs of parallel sides.

12 Every parallelogram is a rectangle.

13 A rhombus has 4 right angles.

14 Every rectangle is a square.

15 Every square is a rhombus.

16 A kite is a parallelogram.

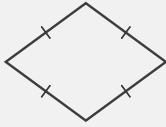
Level 2 · Building up

Medium · 15 questions

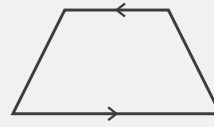
WE DO With your teacher: copy what they write on the board.

Example 2 Name each quadrilateral.

a Name this quadrilateral.



b Name this quadrilateral.



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

15 QUESTIONS

REMEMBER **RHOMBUS**: parallelogram with all sides equal. **RECTANGLE**: parallelogram with a right angle.

ROUND 1 Questions 1–7: how many pairs of parallel sides? How many right angles?

1 isosceles trapezium

2 square

3 rectangle

4 parallelogram

5 trapezium

6 kite

7 rhombus

YOU DO

On your own: **Medium** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2

Questions 8–15: name the most specific quadrilateral.

- | | |
|---|---|
| 8 a quadrilateral with only one diagonal as an axis of symmetry | 9 a quadrilateral with 2 pairs of parallel sides and equal diagonals |
| 10 a quadrilateral whose diagonals bisect each other | 11 a quadrilateral whose diagonals are equal and bisect each other at 90° |
| 12 a quadrilateral whose diagonals bisect each other at right angles | 13 a kite with 4 equal sides |
| 14 a quadrilateral with one pair of parallel sides and equal legs | 15 a rhombus with a right angle |



Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Give the definitions.

- a** What is the definition of a rhombus? **b** What am I? I have one pair of parallel sides.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Definitions.

3 QUESTIONS

REMEMBER **SQUARE**: rectangle with all sides equal. **KITE**: two pairs of equal adjacent sides.

- 1 What quadrilateral am I? I have both pairs of opposite sides parallel.
 - 2 Write the definition of a rectangle.
 - 3 Is a square a rhombus? Explain using definitions.



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Draw a family tree of the quadrilaterals.

1. Quadrilateral → trapezium → parallelogram
2. Parallelogram → rectangle and rhombus
3. Rectangle and rhombus → square

Answer: A square is a special rectangle and a special rhombus

YOU DO

On your own: read the worked example, then try these.

- E1** Where does a kite fit in the family tree? Is a rhombus a kite?

- E2** Which special quadrilaterals have exactly 2 axes of symmetry?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Is a square convex?

Level 2

Name a quadrilateral with exactly one pair of parallel sides.

Level 3

What is the definition of a parallelogram?

Lesson summary

Missed the lesson? Start here.

Definitions

1. **Trapezium:** at least one pair of parallel sides.
Parallelogram: both pairs parallel.
2. **Rhombus:** parallelogram with all sides equal.
Rectangle: parallelogram with a right angle.
3. **Square:** rectangle with all sides equal. **Kite:** two pairs of equal adjacent sides.
4. **Convex:** all angles less than 180° .

Level 1

Dart shape

1. One angle is reflex

Answer: non-convex

Level 2

Opposite sides parallel

1. Both pairs

Answer: parallelogram

Level 3

Rhombus definition

1. Parallelogram, all sides equal

Answer: a parallelogram with 4 equal sides

Extension

Square in the family

1. Rectangle and rhombus

Answer: both

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
6.09

Angle sum of a quadrilateral

Date _____

Learning intentions

We are learning to:

- use the angle sum of a quadrilateral to find unknown angles

Success criteria

I can:

- ☐ know the angles in a quadrilateral add to 360°
- ☐ find an unknown angle
- ☐ use right angles and equal angles in special quadrilaterals

Key terms Write the definition as your teacher explains it.

Angle sum

Quadrilateral

Notes

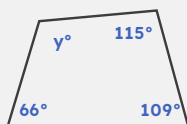
Level 1 · Getting started

Easy · 21 questions

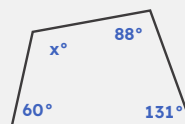
WE DO With your teacher: copy what they write on the board.

Example 1 Find the unknown angle. Angles in a quadrilateral add to 360° .

a Find y .



b Find x .



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **21 QUESTIONS**

REMEMBER Angles in a quadrilateral add to 360° .

ROUND 1 Questions 1–9: three angles of a quadrilateral are given. Find the fourth. (Angles add to 360° .)

1 $126^\circ, 108^\circ, 72^\circ$

2 $104^\circ, 139^\circ, 77^\circ$

3 $93^\circ, 100^\circ, 91^\circ$

4 $139^\circ, 112^\circ, 50^\circ$

5 $96^\circ, 64^\circ, 94^\circ$

6 $97^\circ, 135^\circ, 65^\circ$

7 $124^\circ, 104^\circ, 52^\circ$

8 $118^\circ, 106^\circ, 80^\circ$

9 $71^\circ, 57^\circ, 88^\circ$

YOU DO

On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 10–21: a parallelogram has this angle. Find the other three angles.

10 111°

11 26°

12 112°

13 42°

14 119°

15 23°

16 31°

17 146°

18 27°

19 36°

20 106°

21 160°

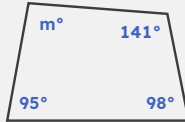
Level 2 · Building up

Medium · 24 questions

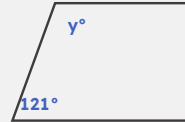
WE DO With your teacher: copy what they write on the board.

Example 2 Find the unknown angle.

a Find m .



b Find y .



Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS**

REMEMBER A diagonal makes 2 triangles: $2 \times 180^\circ$.

ROUND 1 Questions 1–12: a kite has two equal opposite angles, x . Find x .

1 other angles 82° and 84°

2 other angles 52° and 52°

3 other angles 96° and 58°

4 other angles 77° and 65°

5 other angles 44° and 146°

6 other angles 116° and 90°

7 other angles 94° and 134°

8 other angles 85° and 123°

9 other angles 63° and 97°

10 other angles 43° and 59°

11 other angles 58° and 74°

12 other angles 62° and 106°

YOU DO

On your own: **Medium** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 13–24: the angles of a quadrilateral are given. Find x .

13 $x^\circ, 3x^\circ, 3x^\circ, 10^\circ$

14 $2x + 10^\circ, x + 15^\circ, 3x^\circ, 191^\circ$

15 $2x^\circ, x + 10^\circ, 2x + 10^\circ, 75^\circ$

16 $x + 20^\circ, 2x^\circ, 3x^\circ, 280^\circ$

17 $x^\circ, 3x^\circ, x^\circ, 105^\circ$

18 $x^\circ, 3x^\circ, 3x^\circ, 199^\circ$

19 $2x^\circ, 2x^\circ, x^\circ, 285^\circ$

20 $x + 5^\circ, 2x + 5^\circ, 2x^\circ, 215^\circ$

21 $2x^\circ, 2x^\circ, 3x^\circ, 157^\circ$

22 $2x + 15^\circ, 3x^\circ, x^\circ, 105^\circ$

23 $x^\circ, x^\circ, x^\circ, 252^\circ$

24 $2x + 10^\circ, x + 15^\circ, x^\circ, 19^\circ$



Level 3 • Challenge

Challenging · 3 questions · then try the extension

WE DO

With your teacher: copy what they write on the board.

Example 3

Explain, then find the angles.

a Why do the angles in a quadrilateral add to 360° ?

b Three angles of a quadrilateral are equal and the fourth is 90° . Find the equal angles.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER Add the known angles, subtract from 360° .

1 The angles of a quadrilateral are x° , $2x^\circ$, $3x^\circ$ and $4x^\circ$. Find each angle.

2 Can a quadrilateral have 4 obtuse angles? Explain.

3 A parallelogram has one angle of 58° . Find the other three.

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

Find the angle sum of a pentagon by splitting it into triangles.

1. Draw diagonals from one vertex: 3 triangles
2. $3 \times 180^\circ$

Answer: 540°

YOU DO

On your own: read the worked example, then try these.

- E1 Find the angle sum of a hexagon.

- E2 Find each angle of a regular pentagon.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Three angles are 90° , 90° and 70° . Find the fourth.

Level 2

Three angles are 100° , 85° and 95° . Find the fourth.

Level 3

A parallelogram has an angle of 65° . Find the other angles.

Lesson summary

Missed the lesson? Start here.

- Remember**
- Angles in a quadrilateral add to 360° .
 - A diagonal makes 2 triangles: $2 \times 180^\circ$.
 - Add the known angles, subtract from 360° .
 - Use properties: opposite angles in a parallelogram are equal.

Level 1

$66^\circ, 109^\circ, 115^\circ, y^\circ$

1. $360 - 290$

Answer: $y = 70$

Level 2

$60^\circ, 131^\circ, 88^\circ, x^\circ$

1. $360 - 279$

Answer: $x = 81$

Level 3

Two right angles and 121°

1. $360 - 90 - 90 - 121$

Answer: $y = 59$

Extension

Pentagon

1. 3 triangles

Answer: 540°

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON 6.10

Properties of quadrilaterals

Date _____

Learning intentions

We are learning to:

- describe the side, angle and diagonal properties of special quadrilaterals

Success criteria

I can:

- ☐ list the properties of a parallelogram
- ☐ describe how the diagonals behave in each special quadrilateral
- ☐ identify a quadrilateral from its properties

Key terms Write the definition as your teacher explains it.

Diagonal

Bisect

Notes

Level 1 · Getting started

Easy · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the properties of a parallelogram.

- a** List 2 properties of a parallelogram.

Your teacher works this out. Copy every step.

- b** EFGH is a parallelogram. Which is NOT true? A $\angle EFG = \angle EHG$ B $EF = HG$ C $EF = FG$ D $\angle EFG + \angle FGH = 180^\circ$

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

12 QUESTIONS

12 QUESTIONS

REMEMBER Parallelogram: diagonals **BISECT** each other.

ROUND 1 Questions 1–6: true or false?

- 1

The diagonals of a rhombus bisect its angles.

2

Co-interior angles in a parallelogram add to 180° .

3

The opposite sides of a parallelogram are equal.

4

A rhombus has equal diagonals.

5

The adjacent sides of a rectangle are always equal.

6

A square is a special kite.

YOU DO

On your own: **Easy** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2

Questions 7–12: which quadrilaterals have this property? Name all you know.

7 diagonals meet at 90°

8 diagonals bisect each other

9 all sides equal

10 diagonals are equal

11 opposite angles equal

12 exactly one pair of parallel sides

 Level 2 · Building up

Medium · 18 questions

WE DO With your teacher: copy what they write on the board.**Example 2** Name all the quadrilaterals that fit.**a** What am I? All my angles are equal.

Your teacher works this out. Copy every step.

b What am I? My diagonals are equal and bisect each other.

Your teacher works this out. Copy every step.

YOU DOOn your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

18 QUESTIONS

REMEMBER Rectangle: diagonals equal and bisect.**ROUND 1** Questions 1–12: use the properties to find the missing value.**1** rhombus: one angle is 50° . The opposite angle?

2 parallelogram: sides 9 cm and 5 cm. The perimeter?

3 kite: angles 100° , 80° , 80° . The fourth?

4 isosceles trapezium: one base angle is 65° . The other base angle?

5 rectangle: one diagonal is 13 cm. The other?

6 parallelogram: one angle is 70° . The angle next to it?

7 rectangle: half a diagonal is 6 cm. The whole diagonal?

8 square: diagonals meet at what angle?

9 parallelogram: one angle is 125° . The opposite angle?

10 rhombus: one side is 7 cm. The perimeter?

11 square: perimeter 48 cm. The side?

12 rhombus: perimeter 36 cm. One side?

YOU DO On your own: **Medium** keep going: Round 2.

6 QUESTIONS MORE

ROUND 2 Questions 13–18: give a reason: why is each statement true?**13** A square is a rhombus.**14** A rectangle is a parallelogram.**15** A rhombus is a parallelogram.**16** A square is a kite.**17** A square is a rectangle.**18** A rhombus is a kite.**Level 3 • Challenge**

Challenging • 3 questions • then try the extension

WE DO With your teacher: copy what they write on the board.**Example 3** Name all the quadrilaterals that fit.**a** What am I? My diagonals bisect each other.

Your teacher works this out. Copy every step.

b What am I? My opposite angles are equal.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Use properties.

3 QUESTIONS

REMEMBER Rhombus: diagonals bisect at 90° and bisect the angles.**1** ABCD is a parallelogram with $AB = 7$ cm, $BC = 4$ cm and $\angle A = 65^\circ$. Find CD, AD and $\angle C$.**2** The diagonals of a rhombus are 6 cm and 8 cm. What angle do they meet at?**3** What am I? My diagonals are equal, bisect each other and are perpendicular.



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

A rhombus has diagonals 10 cm and 24 cm.
Find its side length.

1. Diagonals bisect each other at right angles:
halves are 5 and 12
2. $\text{Side}^2 = 5^2 + 12^2 = 169$ (Pythagoras, from Stage 5)

Answer: 13 cm

YOU DO

On your own: read the worked example,
then try these.

- E1 A rectangle has sides 6 cm and 8 cm. Its diagonals are 10 cm. How far is the centre from each corner?

- E2 Explain why a square's diagonals make angles of 45° with its sides.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show
your working.

Name _____ Class _____

Level 1

Name a quadrilateral with equal diagonals.

Level 2

State one property of the diagonals of a rhombus.

Level 3

What am I? My opposite sides are equal and all angles are 90° .

Lesson summary

Missed the lesson? Start here.

- Diagonals**
1. Parallelogram: diagonals **bisect** each other.
 2. Rectangle: diagonals equal and bisect.
 3. Rhombus: diagonals bisect at 90° and bisect the angles.
 4. Square: all of these. Kite: one diagonal bisects the other at 90° .

Level 1

Parallelogram properties

1. Opposite sides equal
2. Opposite angles equal

Answer: any 2

Level 2

EFGH: which is not true?

1. Adjacent sides need not be equal

Answer: C

Level 3

Diagonals equal and bisect

1. Rectangle family

Answer: rectangle, square

Extension

Rhombus diagonals 10, 24

1. Halves 5, 12

Answer: 13 cm

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.